

A 2-sectors, 2-factors general equilibrium model

Introduction to CGE Modeling Training

Arief Anshory Yusuf

Problem

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- In a small island faraway from everywhere, there are only two activities: farming (growing corns) and domesticating cattle (livestock). Define farming as X_1 and livestock as X_2 . Both activities need two resources: land and labor. Define land as K and labor as L .
- The island is facing climate change threat. In the near future, sea level rise will reduce the land area by half (50 percent).
- In order to anticipate this, the village want to know how much the harvest of corn will change compared to the current harvest (in percent) and how much meat production will also change relative to the current production (in percent). In short:
 - **How much corn production (X_1) and meat production (X_2) will change (in %) as a result of 25 percent decline in land area due to sea level rise?**

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The noble and intellectually-gifted chief of the tribe observed (historically) the following:

- If both land area and number of labor are increased at the same time by 10% to farming, the corn harvest increased by 10% too. The same thing happened to cattle domesticating activities. Their output always increase by the same proportion with the increase of all their inputs.
- However, when land area is increased to farming by 10%, without increasing labor, harvest increased by only 6%.
- The chief also observed that when 10% more land is allocated to livestock (no labor is increased), production increases by only 3%.

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The meticulously-observing chief also remembers that:

- During the tribe's festivities, people are a lot more happier when more meat are served than when more corn is served with the same proportional change.
- The tribe's chief then made an educated guess that when meat consumption increases by 10%, his people are 7% happier, and when corn consumption increases by 10%, his people are 3% happier.

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So, to the question: **How much corn production (X_1) and meat production (X_2) will change (in %) as a result of 50 percent decline in land area due to sea level rise?** The answer is:

- Corn production (X_1) will decrease by 34.0%
- Meat production (X_2) will decrease by 18.8%

The solution's formula

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The formula to come up with such answer is the following:

$$x_1 = 100 \left[\left(\frac{100+k_s}{100} \right)^\alpha \left(\frac{100+l_s}{100} \right)^{1-\alpha} - 1 \right] = 100 \left[\left(\frac{100-50}{100} \right)^{0.6} \left(\frac{100+0}{100} \right)^{0.4} - 1 \right] = 34.0$$
$$x_2 = 100 \left[\left(\frac{100+k_s}{100} \right)^\beta \left(\frac{100+l_s}{100} \right)^{1-\beta} - 1 \right] = 100 \left[\left(\frac{100-50}{100} \right)^{0.3} \left(\frac{100+0}{100} \right)^{0.7} - 1 \right] = 18.8$$

where x_1 is the percentage change increase in X_1 (corn), x_2 is the percentage change increase in X_2 (meat), $k_s = -50$ is the percentage change increase in land area available (K_s), $l_s = 0$ is the percentage change increase in labor supply available (L_s), $\alpha = 0.6$ is the elasticity of farm output with respect to land input (K_1), while $\beta = 0.3$ is the elasticity of livestock output with respect to land input (K_2).

Level vs percentage change

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- Recall that the question is: **How much corn production (X_1) and meat production (X_2) will change (in %) as a result of 50 percent decline in land area due to sea level rise?**
- What if the question is slightly changed: **What will be the new level of corn production (X_1) and meat production (X_2) as a result of 50% decline in land area due to sea level rise?**
- First and foremost we have to understand how the level of production of X_1 and X_2 is decided at the first place (existing level of production).
 - To do this, first, we will make assumption that the tribe's chief is the one who decide the level of production of the two product, and by doing so, allocating how much land and labor is allocated to the two activities.
 - The interest of the chief is one and only one: to make sure its people gets the greatest satisfaction (happiness/utility) from the consumption of the two product.

The level solution is a bit more complex

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The level solution can be found if we know the level of available land area in the island (K_S) and the number of worker available (L_S), then use the following formula:

$$X_1 = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\alpha \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\alpha}$$

$$X_2 = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\beta \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\beta}$$

where X_1 and X_2 is the level (quantity) of corn harvest and meat production respectively, γ is the elasticity of happiness/utility with respect to corn consumption, K_S is the available land area in the island, and L_S is the number of workers in the island. **Where this formula come from?**

Remember the setup: Technology

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- If both land and labor are increased at the same time by 10 pct to farming, the corn harvest increased by 10 pct too. The same thing happened to cattle domesticating activities. Their output always increase by the same proportion with the increase of all their inputs. **Constant returns to scale/CRS**
- However, when land area is increased to farming by 10 pct, without increasing labor, harvest increased by only 6 pct. **Output elasticity to land, $\alpha = 0.6$.**
- The chief also observed that when 10 pct more land is allocated to livestock (no labor is increased), production increases by only 3 pct. **Output elasticity to land, $\beta = 0.3$.**
- Therefore: $X_1 = K_1^\alpha L_1^{1-\alpha}$, $X_2 = K_2^\beta L_2^{1-\beta}$, Cobb-Douglas production function with CRS or Constant Elasticity of Substitution (CES) with unit elasticity of substitution.

Remember the setup: preferences

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- During the tribe's festivities, people are a lot more happier when more meat are served than when more corn is served with the same proportional change.
- The tribe's chief then made an educated guess that when meat consumption increases by 10 pct, his people (more often than not) are 7 pct happier, and when corn consumption increases by 10 pct, his people are 3 pct happier.
- This can be represented by a Cobb-Douglas utility function $U = X_1^\gamma X_2^{1-\gamma}$ where $\gamma = 0.3$

The Chief of tribe's problem

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The tribe's chief tries to find the best allocation of resource i.e., K_1 , K_2 , L_1 and L_2 that maximizes the tribe's utility (U):

$$U = X_1^\gamma X_2^{1-\gamma}$$

subject to technology and endowment constraints:

$$X_1 = K_1^\alpha L_1^{1-\alpha}, X_2 = K_2^\beta L_2^{1-\beta}$$

$$K_1 + K_2 = K_S, L_1 + L_2 = L_S$$

We can solve this problem mathematically by substituting constraints equations into utility function, Then use calculus of (unconstrained) optimization to find the optimal resource allocation solution for K_1 , K_2 , L_1 and L_2 . **See derivation in the appendix at the end of this slide.**

Applying these formula

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Applying these formula and knowing that $\alpha = 0.6$, $\beta = 0.3$, $\gamma = 0.3$, $K_S = 100$ and $L_S = 100$:

$$X_1 = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\alpha \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\alpha} = 32.8$$

$$X_2 = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\beta \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\beta} = 71.2$$

If you apply new land area $K_S = 50$, then you have:

$$X_1 = 21.6 \text{ and } X_2 = 57.9$$

It's a decrease by 34.0% and 18.8% respectively.

Diagrammatic illustration in Excel

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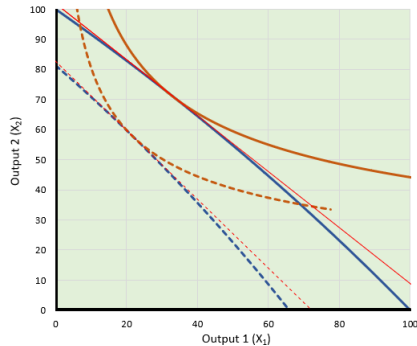
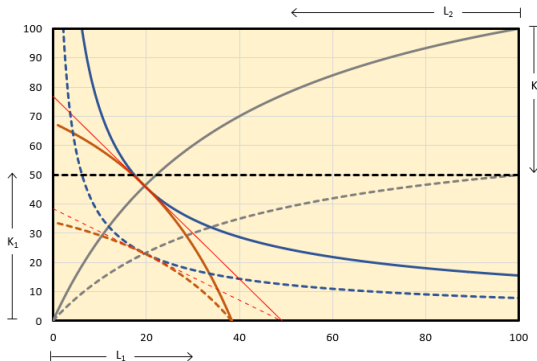
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- You can open excelppf.xlsx to play around.

From level to (exact, not approximate) percent change solution

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From level solution:

$$X_1 = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\alpha \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\alpha}, \text{ and}$$
$$X_2 = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\beta \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\beta}$$

to percentage change solution:

$$x_1 = 100 \left[\left(\frac{100+k_s}{100} \right)^\alpha \left(\frac{100+l_s}{100} \right)^{1-\alpha} - 1 \right], \text{ and}$$
$$x_2 = 100 \left[\left(\frac{100+k_s}{100} \right)^\beta \left(\frac{100+l_s}{100} \right)^{1-\beta} - 1 \right]$$

Try to do derive it yourself!

Social planner v.s. competitive market solution

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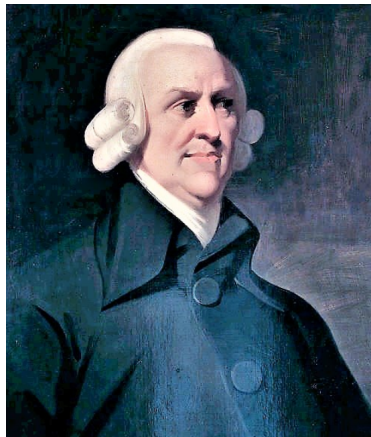
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Noble and intelligent tribe's chief is rare.
The reality is market system.

“It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own self-interest. We address ourselves not to their humanity but to their self-love, and never talk to them of our own necessities, but of their advantages”

Adam Smith: The Wealth of Nations



Competitive market setup: everyone think of him/herself

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We introduce P_1 (price of corn), P_2 (price of meat), W (wage) and R (land rent) and distinguish supply (X) and demand (Q)

- Producer's problem:
 - Producer 1: $\min WL_1 + RK_1$ subject to $X_1 = K_1^\alpha L_1^{1-\alpha}$
 - Producer 2: $\min WL_2 + RK_2$ subject to $X_2 = K_2^\beta L_2^{1-\beta}$
- Consumer's problem:
 - $\max U = Q_1^\gamma Q_2^{1-\gamma}$ subject to $Y = P_1 Q_1 + P_2 Q_2$, Y is income.
- Solution to producer's problem (**ask for handsout!**):
 - Producer 1: $K_1 = \frac{\alpha c_1 X_1}{R}$; $L_1 = \frac{(1-\alpha)c_1 X_1}{W}$
 - Producer 2: $K_2 = \frac{\beta c_2 X_2}{R}$; $L_2 = \frac{(1-\beta)c_2 X_2}{W}$
 - Competitive market implies zero profit: $P_1 = c_1$; $P_2 = c_2$, c is unit cost.
- Solution to consumer's problem: $Q_1 = \gamma \frac{Y}{P_1}$; $Q_2 = (1-\gamma) \frac{Y}{P_2}$

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- Demand for land

- $K_1 = \frac{\alpha c_1 X_1}{R}$

- $K_2 = \frac{\beta c_2 X_2}{R}$

- Demand for labor

- $L_1 = \frac{(1-\alpha)c_1 X_1}{W}$

- $L_2 = \frac{(1-\beta)c_2 X_2}{W}$

- Demand for corns and meats

- $Q_1 = \gamma \frac{Y}{P_1}$

- $Q_2 = (1 - \gamma) \frac{Y}{P_2}$

- Zero profit conditions

- $P_1 = c_1$

- $P_2 = c_2$

- Market clearing (equilibrium) conditions for commodities

- $Q_1 = X_1$

- $Q_2 = X_2$

- Market clearing (equilibrium) conditions for inputs

- $K_1 + K_2 = K_S$

- $L_1 + L_2 = L_S$

By substitutions and some algebraic manipulation, it turns out that the solution is identical to social planner (noble tribe's chief) solution (See appendix for derivation).

Competitive solution: Factor allocation

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The equilibrium for factor (labor and land) allocations:

$$K_1^e = \frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S$$

$$K_2^e = \frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S$$

$$L_1^e = \frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S$$

$$L_2^e = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S$$

They depend on (1) factor endowment, (2) technology, and (3) preference *And they are exactly the same as the social-planner solution!*

Competitive solution: Output

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Substitute optimal factor allocation into production function:

$$X_1 = K_1^\alpha L_1^{1-\alpha}, X_2 = K_2^\beta L_2^{1-\beta}$$

$$X_1^e = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\alpha \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\alpha}$$

$$X_2^e = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\beta \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\beta}$$

Competitive solution: Prices

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There is no central bank in the island. It's a “corn” economy. We can use one commodity/factor as a *numeraire*. Suppose corn is the unit of exchange, so everything is valued with corns then P_1 is that *numeraire*. Then all prices are relative to P_1 . The competitive solutions of prices become:

$$P_1 = 1 \quad P_2 = \frac{1 - \gamma}{\gamma} \frac{X_1^e}{X_2^e} \quad W = \frac{(1 - \beta) X_1^e}{L_1^e} \quad R = \frac{\beta X_1^e}{K_1^e}$$

Note: Use equations in optimum solutions to define prices.

Competitive solution: Walras Law

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- Walras' law is a principle in general equilibrium theory asserting that budget constraints imply that the values of excess demand (or, conversely, excess market supplies) must sum to zero.
- It implies that when there is N market in the economy and the $N - 1$ market is in equilibrium, then the $N - th$ market must be in equilibrium.
- Nominal homogeneity $\Rightarrow$ if all prices double, no change in quantities or real variables
- Only relative prices matters.
- **It implies that we must remove one of the (market clearing) equations in the CGE models**

Our general equilibrium equations, with $i = 1, 2$

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- ❶ $K_i = \frac{\alpha_i c_i X_i}{R} \Leftarrow$ Demand for capital
- ❷ $L_i = \frac{(1-\alpha_i) c_i X_i}{W} \Leftarrow$ Demand for labor
- ❸ $c_i = \left(\frac{R}{\alpha_i}\right)^{\alpha_i} \left(\frac{W}{1-\alpha_i}\right)^{1-\alpha_i} \Leftarrow$ Unit cost definition
- ❹ $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit
- ❺ $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities
- ❻ $Y = W \cdot L_S + R \cdot K_S \Leftarrow$ Household income
- ❼ $K_S = \sum_i K_i \Leftarrow$ Market clearing for capital
- ❽ $L_S = \sum_i L_i \Leftarrow$ Market clearing for Labor
- ❾ $Q_i = X_i \Leftarrow$ Market clearing for commodities

Counting variables and equations

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- From equation 1 to 9, and $i = 1, 2$, we have
 - 15 equations, to determine
 - 17 variables: $K_1, L_1, X_1, K_2, L_2, X_2, Q_1, Q_2, c_1, c_2, P_1, P_2, W, R, Y, K^S, L^S$
- To close the system, by Walras' law one market clearing will be dropped. It gives now:
 - 14 equations, to determine
 - 14 variables, because K^S, L^S are exogenous, and one of the price, let's say P_1 will be used as numeraire.

With CES technology/production function

Ask for handout!

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❶ $K_i = X_i \alpha_i^{\sigma_i} \left(\frac{R}{c_i} \right)^{-\sigma_i} \Leftarrow$ Demand for capital

❷ $L_i = X_i \beta_i^{\sigma_i} \left(\frac{W}{c_i} \right)^{-\sigma_i} \Leftarrow$ Demand for labor

❸ $c_i = (\alpha_i^{\sigma_i} R^{1-\sigma_i} + \beta_i^{\sigma_i} W^{1-\sigma_i})^{\frac{1}{1-\sigma_i}} \Leftarrow$ Unit cost definition

❹ $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit

❺ $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities

❻ $Y = W \cdot L^S + R \cdot K^S \Leftarrow$ Household income

❼ $K^S = \sum_i K_i \Leftarrow$ Market clearing for capital

❽ $L^S = \sum_i L_i \Leftarrow$ Market clearing for Labor

❾ $Q_i = X_i \Leftarrow$ Market clearing for commodities

Model's parameterization/Calibration

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- Data (of the economy) represent initial equilibrium: *Important Assumption!!*
- Input-Output Table of the island's economy:

	Industry		Consumption	Total
	1-corn	2-meat		
1-corn			$P_1 X_1$	$P_1 X_1$
2-meat			$P_2 X_2$	$P_2 X_2$
Salary/labor cost	WL_1	WL_2		WL^S
Surplus/land rental	RK_1	RK_2		RK^S
Total	$P_1 X_1$	$P_2 X_2$		Y

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- I-O Table of the island's economy

	Industry		Consumption	
	1-corn	2-meat		Total
1-corn			100	100
2-meat			200	200
Salary/labor cost	40	140		180
Surplus/land rental	60	60		120
Total	100	200	300	300

- Calibration generally means "finding parameters of GE model (i.e. α, β, γ) that can reproduce the initial equilibrium exactly like this data".

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- From Input-Output Table, $WL_1 = 40$, $WL_2 = 140$, $RK_1 = 60$, and $RK_2 = 60$.
- Then, $P_1X_1 = WL_1 + RK_1 = 100$, and $P_2X_2 = WL_2 + RK_2 = 200$.
- Then, derived from first order conditions

$$\alpha = \frac{RK_1}{P_1X_1} = \frac{60}{100} = 0.6$$

$$\beta = \frac{RK_2}{P_2X_2} = \frac{60}{200} = 0.3$$

$$\gamma = \frac{P_1X_1}{P_1X_1 + P_2X_2} = \frac{100}{300} = 0.3$$

- Now, we have all the parameters!

Typical steps in the CGE modeling process with Gempack

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- 1 Understand the theory (in level equations)
- 2 Linearization of the level equations
- 3 Getting and prepare the data (compiling in Excel)
- 4 Creating Tablo file (of linearized model's equations)
- 5 Creating database file (Moving from Excel to HAR files)
- 6 Preparing simulation and simulation file
- 7 Running the simulation and analyze the results

Step 1 - Understanding The theory (level equations)

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❸ $c_i = (\alpha_i^{\sigma_i} R^{1-\sigma_i} + \beta_i^{\sigma_i} W^{1-\sigma_i})^{\frac{1}{1-\sigma_i}} \Leftarrow$ Unit cost definition

❹ $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit

❺ $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities

❻ $Y = W \cdot L^S + R \cdot K^S \Leftarrow$ Household income

❼ $L^S = \sum_i L_i \Leftarrow$ Market clearing for Labor

❽ $Q_i = X_i \Leftarrow$ Market clearing for commodities

Notes that market clearing for capital is removed by Walras' law?

Step 2. Linearization (percentage change)

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- ① $k_i = x_i - \sigma_i (r - c_i) \Leftarrow$ Demand for capital
- ② $l_i = x_i - \sigma_i (w - c_i) \Leftarrow$ Demand for labor
- ③ $c_i = S_i^L \cdot w + (1 - S_i^L) \cdot r \Leftarrow$ Unit cost definition
- ④ $P_i X_i \cdot (p_i + x_i) = WL_i \cdot (w + l_i) + RK_i (r + k_i) \Leftarrow$ zero profit
- ⑤ $q_i = y - p_i \Leftarrow$ Demand for commodities
- ⑥ $Y \cdot y = WL^S \cdot (w + l^S) + RK^S \cdot (r + k^S) \Leftarrow$ Household income
- ⑦ $x_i = q_i \Leftarrow$ Market clearing for commodities
- ⑧ $WL^S \cdot l^S = \sum_i WL_i \cdot l_i \Leftarrow$ Market clearing for Labor

Step 3. Getting and preparing the data

Input-Output table:

	Industry		Consumption	
	1-corn	2-meat		Total
1-corn			100	100
2-meat			200	200
Salary/labor cost	40	140		180
Surplus/land rental	60	60		120
Total	100	200	300	300

- WL_i and RK_i are base coefficients.
- Other coefficients, $P_i X_i = WL_i + RK_i$, $WL^S = \sum_i WL_i$, $RK^S = \sum_i RK_i$, $Y = WL^S + RK^S$ are derived from base coefficients.

Step 4. Creating Tablo File of model's linearized equations

Typical model's file structure

- 1 FILE statement. Declaring a 'logical' file name where your data will be located.
- 2 SET statement. Declaring set names, or subscript in the model.
- 3 VARIABLE statement. Declaring all the variables in the model.
- 4 COEFFICIENT statement. Declaring all the coefficient and parameter in the model.
- 5 READ statement. Asking the program to read the 'base' coefficient from database.
- 6 FORMULA statement. Calculation of other coefficient as a function of base coefficient.
- 7 UPDATE statement. Updating the base coefficient.
- 8 EQUATION statement. Listing all the model equations.

Step 5. Creating Database File

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- Database file record the 'base' coefficient, here the coefficients are WLi and RK_i .
- In GEMPACK the database file has the extension .har, and created and read by a GEMPAK program called VIEWHAR.

Step 6. Creating Simulation File

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- Simulation file is a text file with extension .CMF where you put the closure and shock statement, and some other ancilliary information.
- *Closure* is a statement declaring which variables are endogenous, and which variables are exogenous that make sure that the number of endogenous variables is equal to the number of statement and also will tell us what variables that can be shocked.
- *Shock statement* is changing one or combination of exogenous variables (our policy question) to see its impact on all the endogenous variables.
- other information include: solution methods, location of physical (not logical) database and output, etc.

Step 7. Running the simulation and analyze the results

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Asking GEMPACK to implement our 6 steps. Now, it is your first time for computer exercise with Gempack!!

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A benevolent social planner maximizes $U = X_1^\gamma X_2^{1-\gamma}$
subject to $X_1 = K_1^\alpha L_1^{1-\alpha}$, $X_2 = K_2^\beta L_2^{1-\beta}$, $K_1 + K_2 = K_S$, $L_1 + L_2 = L_S$.

- Note that no prices here!

By substitution $X_2 = (K_S - K_1)^\beta (L_S - L_1)^{1-\beta}$, then

$$U = (K_1^\alpha L_1^{1-\alpha})^\gamma ((K_S - K_1)^\beta (L_S - L_1)^{1-\beta})^{1-\gamma}, \text{ or}$$

$$U = K_1^{\alpha\gamma} (K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma} (L_S - L_1)^{(1-\beta)(1-\gamma)}$$

First order conditions of maximization:

$$\frac{\partial U}{\partial K_1} = 0 \text{ and } \frac{\partial U}{\partial L_1} = 0$$

Social planner's solution (cont.)

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$$U = K_1^{\alpha\gamma} (K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma} (L_S - L_1)^{(1-\beta)(1-\gamma)}$$

$$\frac{\partial U}{\partial K_1} = 0 \Rightarrow \alpha\gamma K_1^{\alpha\gamma-1} (K_S - K_1)^{\beta(1-\gamma)} - K_1^{\alpha\gamma} \beta(1-\gamma) (K_S - K_1)^{\beta(1-\gamma)-1} = 0$$

$$\alpha\gamma K_1^{\alpha\gamma-1} (K_S - K_1)^{\beta(1-\gamma)} = K_1^{\alpha\gamma} \beta(1-\gamma) (K_S - K_1)^{\beta(1-\gamma)-1}$$

$$\alpha\gamma K_1^{-1} = \beta(1-\gamma) (K_S - K_1)^{-1} \Rightarrow \frac{\alpha\gamma}{K_1} = \frac{\beta(1-\gamma)}{K_S - K_1} \Rightarrow \frac{K_S - K_1}{K_1} = \frac{\beta(1-\gamma)}{\alpha\gamma}$$

$$\boxed{K_1^e = \frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S} \quad \text{and} \quad \boxed{K_2^e = \frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S}$$

Social planner's solution (cont.)

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$$U = K_1^{\alpha\gamma} (K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma} (L_S - L_1)^{(1-\beta)(1-\gamma)} \Rightarrow \frac{\partial U}{\partial L_1} = 0$$

$$(1-\alpha)\gamma L_1^{(1-\alpha)\gamma-1} (L_S - L_1)^{(1-\beta)(1-\gamma)} - L_1^{(1-\alpha)\gamma} (1-\beta)(1-\gamma) (L_S - L_1)^{(1-\beta)(1-\gamma)-1} = 0$$

$$(1-\alpha)\gamma L_1^{-1} = (1-\beta)(1-\gamma) (L_S - L_1)^{-1} \Rightarrow \frac{(1-\alpha)\gamma}{L_1} = \frac{(1-\beta)(1-\gamma)}{L_S - L_1}$$

$$\Rightarrow \frac{L_S - L_1}{L_1} = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma}$$

$$\boxed{L_1^e = \frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S} \text{ and } \boxed{L_2^e = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S}$$

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- Producer's problem:
 - Producer 1: $\min WL_1 + RK_1$ subject to $X_1 = K_1^\alpha L_1^{1-\alpha}$
 - Producer 2: $\min WL_2 + RK_2$ subject to $X_2 = K_2^\beta L_2^{1-\beta}$
- Consumer's problem:
 - $\max U = X_1^\gamma X_2^{1-\gamma}$ subject to $Y = P_1 X_1 + P_2 X_2$
- Solution to producer's problem (see previous module):
 - Producer 1: $K_1 = \frac{\alpha c_1 X_1}{R}; L_1 = \frac{(1-\alpha)c_1 X_1}{W}$
 - Producer 2: $K_2 = \frac{\beta c_2 X_2}{R}; L_2 = \frac{(1-\beta)c_2 X_2}{W}$
 - Competitive market implies zero profit: $P_1 = c_1; P_2 = c_2$
- Solution to consumer's problem:
 - $X_1 = \gamma \frac{Y}{P_1}; X_2 = (1-\gamma) \frac{Y}{P_2} \Rightarrow \frac{P_1 X_1}{P_2 X_2} = \frac{\gamma}{1-\gamma} \Rightarrow P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$

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- Producer 1: $K_1 = \frac{\alpha c_1 X_1}{R} \Rightarrow R = \frac{\alpha c_1 X_1}{K_1}; L_1 = \frac{(1-\alpha)c_1 X_1}{W} \Rightarrow W = \frac{(1-\alpha)c_1 X_1}{L_1}$
- Producer 2: $K_2 = \frac{\beta c_2 X_2}{R} \Rightarrow R = \frac{\beta c_2 X_2}{K_2}; L_2 = \frac{(1-\beta)c_2 X_2}{W} \Rightarrow W = \frac{(1-\beta)c_2 X_2}{L_2}$
- Equate R and use zero profit:
$$\frac{\alpha c_1 X_1}{K_1} = \frac{\beta c_2 X_2}{K_2} \Rightarrow \frac{\alpha P_1 X_1}{K_1} = \frac{\beta P_2 X_2}{K_2}$$
- Substitute this: $P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$
$$\frac{\alpha P_1 X_1}{K_1} = \frac{\beta \frac{1-\gamma}{\gamma} P_1 X_1}{K_2} \Rightarrow \frac{\alpha}{K_1} = \frac{\beta(1-\gamma)}{\gamma K_2}$$
- Equate W and use zero profit:
$$\frac{(1-\alpha)c_1 X_1}{L_1} = \frac{(1-\beta)c_2 X_2}{L_2} \Rightarrow \frac{(1-\alpha)P_1 X_1}{L_1} = \frac{(1-\beta)P_2 X_2}{L_2}$$
- Substitute this: $P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$
$$\frac{(1-\alpha)P_1 X_1}{L_1} = \frac{(1-\beta)\frac{1-\gamma}{\gamma} P_1 X_1}{L_2} \Rightarrow \frac{1-\alpha}{L_1} = \frac{(1-\beta)(1-\gamma)}{\gamma L_2}$$

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- We have found:

$$\frac{\alpha}{K_1} = \frac{\beta(1-\gamma)}{\gamma K_2} \text{ and } \frac{1-\alpha}{L_1} = \frac{(1-\beta)(1-\gamma)}{\gamma L_2}$$

- But we have resource constraint equations:

$$K^S = K_1 + K_2 \text{ and } L^S = L_1 + L_2$$

- Solve for K_1 : $\frac{\alpha}{K_1} = \frac{\beta(1-\gamma)}{\gamma(K^S - K_1)} \Rightarrow \alpha\gamma K^S - \alpha\gamma K_1 = \beta(1-\gamma) K_1$

$$(\beta(1-\gamma) + \alpha\gamma) K_1 = \alpha\gamma K^S \Rightarrow K_1 = \frac{\alpha\gamma}{\beta(1-\gamma) + \alpha\gamma} K^S$$

- Solve for L_1 :

$$\frac{1-\alpha}{L_1} = \frac{(1-\beta)(1-\gamma)}{\gamma(L^S - L_1)} \Rightarrow (1-\alpha)\gamma L^S - (1-\alpha)\gamma L_1 = (1-\beta)(1-\gamma) L_1$$

$$((1-\beta)(1-\gamma) + (1-\alpha)\gamma) L_1 = (1-\alpha)\gamma L^S$$

$$\Rightarrow L_1 = \frac{(1-\alpha)\gamma}{(1-\beta)(1-\gamma) + (1-\alpha)\gamma} L^S$$

Competitive solution: Factor allocation

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The optimum for real (or quantity) of factor allocations

$$K_1^e = \frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \quad K_2^e = \frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S$$

$$L_1^e = \frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \quad L_2^e = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S$$

They depend on (1) factor endowment, (2) technology, and (3) preference *And they are exactly the same as the social-planner solution!*

Competitive solution: Output

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Substitute optimal factor allocation into production function:

$$X_1 = K_1^\alpha L_1^{1-\alpha}, X_2 = K_2^\beta L_2^{1-\beta}$$

$$X_1^e = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\alpha \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\alpha}$$

$$X_2^e = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S \right)^\beta \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S \right)^{1-\beta}$$

Appendix: Cobb-Douglas Cost Minimization

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Define X_i = the quantity of input i , W_i = the price of input i , Q = the quantity of output, α_i = share parameter for input i

Firm's problem is to minimize cost of production subject to a given technology:

$$\min_{X_i} \sum_i W_i X_i \text{ subject to } Q = \prod_i X_i^{\alpha_i}$$

Constant returns to scale $\Rightarrow \sum_i \alpha_i = 1$

Setup the Lagrange equation:

$$\mathcal{L} = \sum_i W_i X_i + \lambda \left\{ Q - \prod_i X_i^{\alpha_i} \right\}$$

First order conditions

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$$\frac{\partial \mathcal{L}}{\partial X_i} = W_i - \lambda \alpha_i X_i^{\alpha_i - 1} \prod_{j \neq i} x_j^{\alpha_j} = W_i - \lambda \alpha_i x_i^{-1} x_i^{\alpha_i} \prod_{j \neq i} x_j^{\alpha_j} = 0$$

$$= W_i - \lambda \frac{\alpha_i}{X_i} \prod_j x_j^{\alpha_j} = W_i - \lambda \frac{\alpha_i}{X_i} Q = 0 \Rightarrow \boxed{W_i = \lambda \frac{\alpha_i}{X_i} Q}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_i} = Q - \prod_i x_i^{\alpha_i} = 0 \Rightarrow \boxed{Q = \prod_i x_i^{\alpha_i}}$$

Solving First Order Conditions: Demand for Inputs

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Ratio of two FOCs: $\frac{W_i}{W_j} = \frac{\alpha_j}{\alpha_i} \frac{X_j}{X_i} \Rightarrow X_i = \frac{\alpha_i X_j W_j}{\alpha_j W_i}$

Substitute into production function:

$$Q = \prod_i X_i^{\alpha_i} = \prod_i \left(\frac{\alpha_i X_j W_j}{\alpha_j W_i} \right)^{\alpha_i} = \prod_i \left(\frac{X_j W_j}{\alpha_j} \right)^{\alpha_i} \prod_i \left(\frac{\alpha_i}{W_i} \right)^{\alpha_i}$$

$$Q = \left(\frac{X_j W_j}{\alpha_j} \right)^{\sum_i \alpha_i} \prod_i \left(\frac{\alpha_i}{W_i} \right)^{\alpha_i} \Rightarrow Q = \frac{X_j W_j}{\alpha_j} \prod_i \left(\frac{\alpha_i}{W_i} \right)^{\alpha_i}$$

$$X_j = \frac{\alpha_j Q}{W_j} \prod_i \left(\frac{W_i}{\alpha_i} \right)^{\alpha_i} \Rightarrow \boxed{X_j = \frac{\alpha_j c Q}{W_j} \text{ and } c = \prod_i \left(\frac{W_i}{\alpha_i} \right)^{\alpha_i}}$$

Cobb Douglas Unit Cost and Cost Share

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Unit cost:

$$\frac{\sum_j W_j X_j}{Q} = \frac{\sum_j W_j \frac{\alpha_j c Q}{W_j}}{Q} = \frac{c Q \sum_j \alpha_j}{Q} = c \Rightarrow \boxed{\frac{\sum_j W_j X_j}{Q} = c = \prod_i \left(\frac{W_i}{\alpha_i} \right)^{\alpha_i}}$$

Cost share:

$$\frac{W_j X_j}{c Q} = \frac{W_j \frac{\alpha_j c Q}{W_j}}{c Q} \Rightarrow \boxed{\frac{W_j X_j}{c Q} = \alpha_j}$$

Appendix: CES Cost Minimization

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Define X_i = the quantity of input i , W_i = the price of input i , Q = the quantity of output, δ_i = share parameter for input i , $-1 < \rho < \infty$ is parameter.

Firm's problem is to minimize cost of production subject to a given technology:

$$\min_{X_i} \sum_i W_i X_i \text{ subject to } Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

Setup the Lagrange equation:

$$\mathcal{L} = \sum_i W_i X_i + \lambda \left\{ Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} \right\}$$

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First order conditions:

$$\frac{\partial \mathcal{L}}{\partial X_i} = 0 \text{ for all } i = 1, \dots, n \text{ and } \frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

$$W_i - \lambda \left(-\frac{1}{\rho} \right) \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}-1} \left(-\rho \delta_i X_i^{-\rho-1} \right) = 0$$

$$Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} = 0$$

Solving first order conditions

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Ratio of two FOCs:

$$\frac{W_i}{W_j} = \frac{\delta_i X_i^{-\rho-1}}{\delta_j X_j^{-\rho-1}}$$

Solve for X_i

$$X_i^{-\rho-1} = \frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \Leftrightarrow X_i = \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{1}{-\rho-1}}$$

$$X_i^{-\rho} = \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{-\rho}{-\rho-1}} \Leftrightarrow X_i^{-\rho} = \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{\rho}{1+\rho}}$$

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Substitution:

$$Q = \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}} = \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} X_j^{\frac{-(\rho+1)\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}}$$

$$Q = X_j \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}}$$

Solving FOCs

Solve for X_j

$$X_j = Q \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \left\{ \sum_i \delta_j^{\frac{\rho}{1+\rho}} \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}}$$

$$X_j = Q \left\{ \delta_j^{\frac{\rho}{1+\rho}} \sum_i \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \delta_j^{\frac{1}{1+\rho}} \left\{ \sum_i \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}}$$

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left\{ W_j^{-\frac{\rho}{1+\rho}} \sum_i \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \sum_i \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}}$$

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \sum_i \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}}$$

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$$X_j = Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{-\rho}} \right\}^{\frac{1}{\rho} \frac{-\rho}{1+\rho}}$$

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$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left\{ \frac{W_j}{\left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}} \right\}^{-\frac{1}{1+\rho}}$$

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}}, \text{ where } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$$

Solution: Factor demand equation

Demand for input X_j

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}}, \text{ where } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$$

$$X_j = Q \delta_j^{\sigma} \left(\frac{W_j}{c} \right)^{-\sigma} \text{ and } c = \left(\sum \delta_i^{\sigma} W_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

where $0 < \sigma = \frac{1}{1+\rho} < \infty$ is the elasticity of substitution

It can be shown that c is the unit cost (or average cost) of production.

CES Unit Cost

A 2-sectors,
2-factors
general
equilibrium
model

AA Yusuf

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Unit cost:

$$\begin{aligned}\frac{\sum_j W_j X_j}{Q} &= \frac{\sum_j W_j Q \delta_j^\sigma \left(\frac{W_j}{c}\right)^{-\sigma}}{Q} = \frac{Q c^\sigma \sum_j W_j \delta_j^\sigma W_j^{-\sigma}}{Q} = c^\sigma \sum_j \delta_j^\sigma W_j^{1-\sigma} \\ &= c^\sigma \left(\sum_j \delta_j^\sigma W_j^{1-\sigma} \right)^{\frac{1-\sigma}{1} \frac{1}{1-\sigma}} = c^\sigma c^{1-\sigma} = c\end{aligned}$$

CES Cost share

A 2-sectors,
2-factors
general
equilibrium
model

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Cost share:

$$S_j = \frac{W_j X_j}{cQ} = \frac{W_j Q \delta_j^\sigma \left(\frac{W_j}{c}\right)^{-\sigma}}{cQ} = \frac{Q c^\sigma W_j \delta_j^\sigma W_j^{1-\sigma}}{cQ} = \frac{\delta_j^\sigma W_j^{1-\sigma}}{c^{1-\sigma}}$$

$$S_j = \frac{\delta_j^\sigma W_j^{1-\sigma}}{(\sum_i \delta_i^\sigma W_i^{1-\sigma})^{\frac{1-\sigma}{1-\sigma}}} \Rightarrow \boxed{S_j = \frac{\delta_j^\sigma W_j^{1-\sigma}}{\sum_i \delta_i^\sigma W_i^{1-\sigma}}}$$

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